

EE3621: Digital Signal Processing

Lecture 16: FIR Filter — Linear-Phase Realization

(III B.Tech EEE)

Comprehensive Lecture Notes & Faculty Textbook Reference

1 Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 05:00 (5 mins):** Motivation: The physical significance of Linear Phase (Phase distortion, dispersion, and waveform preservation).
 - **05:00 – 14:00 (9 mins):** Mathematical foundations: Exact conditions for strict and generalized linear phase (Symmetry and Antisymmetry).
 - **14:00 – 22:00 (8 mins):** The 4 Types of Linear-Phase FIR Filters (Classification, frequency response formulations, and boundary zero properties).
 - **22:00 – 30:00 (8 mins):** Linear-Phase Realization Architecture: Exploiting coefficient symmetry via pre-addition to achieve 50% multiplier reduction.
 - **30:00 – 37:00 (7 mins):** Step-by-step worked numerical examples (Odd vs. Even lengths, Symmetric vs. Antisymmetric).
 - **37:00 – 40:00 (3 mins):** Hardware implementation insights (FPGA DSP slices), pedagogical checkpoints, and university exam review.
-

2 1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Explain** the physical and engineering necessity of linear phase in signal processing systems (biomedical, telecommunications, audio, and radar).
2. **Derive** the mathematical conditions on impulse response $h[n]$ that guarantee pure and generalized linear phase from first principles.
3. **Classify** linear-phase FIR filters into Types I, II, III, and IV, determining their frequency response, group delay, and mandatory boundary zeros.
4. **Formulate** the difference equations and folded signal flow graphs that exploit coefficient symmetry to reduce multiplier count by 50%.
5. **Calculate** exact hardware resource counts (multipliers, adders, delay registers) for arbitrary filter lengths M , both odd and even.

6. **Construct** complete linear-phase direct form structures for symmetric and antisymmetric FIR filters.
 7. **Evaluate** the VLSI, ASIC, and FPGA implementation benefits of dedicated pre-adder architectures in modern DSP hardware.
-

3 2. MOTIVATION & PHYSICAL SIGNIFICANCE OF LINEAR PHASE

3.1 2.1 What is Phase Distortion?

When an electrical signal passes through an LTI system with frequency response $H(e^{j\omega}) = |H(e^{j\omega})|e^{j\theta(\omega)}$, two fundamental modifications occur:

1. **Magnitude Modification:** The amplitude of each spectral component at frequency ω is multiplied by $|H(e^{j\omega})|$.
2. **Phase Shift:** The phase of each spectral component is shifted by $\theta(\omega) = \angle H(e^{j\omega})$.

Recall that a phase shift $\theta(\omega)$ corresponds to a time delay. We define two delay metrics:

- **Phase Delay ($\tau_p(\omega)$):** The time delay experienced by a single, pure sinusoidal carrier:

$$\tau_p(\omega) = -\frac{\theta(\omega)}{\omega} \quad (1)$$

- **Group Delay ($\tau_g(\omega)$):** The time delay experienced by the modulation envelope or wave packet formed by a narrow band of frequencies:

$$\tau_g(\omega) = -\frac{d\theta(\omega)}{d\omega} \quad (2)$$

If $\tau_g(\omega)$ is **not constant** across all frequencies, different frequency components of the input signal will travel through the filter at different speeds! This causes **dispersion** (phase distortion):

- Narrow pulses spread out and smear in time.
- High-frequency details arrive at different moments than low-frequency baselines.
- The time-domain shape of the waveform is severely corrupted, even if $|H(e^{j\omega})| = 1$ (all-pass condition).

3.2 2.2 Why Linear Phase is Essential in Engineering

A filter possesses **Linear Phase** if its phase response is a straight line passing through the origin (or with a $\pm\pi/2$ intercept):

$$\theta(\omega) = -\alpha\omega \implies \tau_p(\omega) = \alpha = \text{constant}, \quad \tau_g(\omega) = \alpha = \text{constant} \quad (3)$$

When a filter has linear phase, **all frequency components undergo the exact same time delay α** . The output is an undistorted replica of the input, shifted purely in time by α samples: $y[n] = x[n - \alpha]$.

Linear phase is strictly mandatory in the following practical domains:

1. **Biomedical Instrumentation (ECG and EEG):** An electrocardiogram (ECG) diagnoses life-threatening arrhythmias and myocardial infarction based on the precise morphological shapes and timing intervals of the P-wave, QRS-complex, and T-wave. A non-linear phase filter alters the peak heights and shifts the ST segment, causing fatal false-positive or false-negative clinical diagnoses.
2. **Digital Telecommunications (QAM, PSK, OFDM):** High-speed digital modems transmit rectangular or raised-cosine pulses. If the channel equalization or pulse-shaping filter has non-linear phase, pulse tails spill into adjacent symbol slots, causing **Inter-Symbol Interference (ISI)** and collapsing the eye diagram.
3. **High-Fidelity Audio Processing:** Human hearing is sensitive to the transient phase alignment of musical instruments (drum transients, piano attacks). Non-linear phase smears transients, dulling the acoustic clarity.
4. **Image Processing & Computer Vision:** 2D edge detection filters must have linear phase (zero phase or symmetric phase); otherwise, edges in images ring asymmetrically, creating artificial ghost outlines.

3.3 2.3 Why FIR Filters are Chosen for Linear Phase

A causal, stable Infinite Impulse Response (IIR) filter **cannot** have exact linear phase. For an IIR filter to have linear phase, its poles would have to occur in mirror-image pairs (p_k and $1/p_k^*$). If a pole is inside the unit circle ($|p_k| < 1$), its partner must lie strictly outside ($|1/p_k^*| > 1$), violating causal BIBO stability.

Therefore, whenever strict linear phase is a non-negotiable system requirement, engineers **must utilize Finite Impulse Response (FIR) filters**.

4 3. MATHEMATICAL CONDITIONS FOR LINEAR PHASE

4.1 3.1 Condition for Pure Linear Phase

Consider a causal FIR filter of length M (order $N = M - 1$) with real impulse response $h[n]$ ($0 \leq n \leq M - 1$):

$$H(e^{j\omega}) = \sum_{n=0}^{M-1} h[n]e^{-j\omega n} = |H(e^{j\omega})|e^{j\theta(\omega)} \quad (4)$$

We require pure linear phase:

$$\theta(\omega) = -\alpha\omega, \quad -\pi \leq \omega \leq \pi \quad (5)$$

where α is a constant delay. Substituting into the frequency response:

$$\sum_{n=0}^{M-1} h[n]e^{-j\omega n} = \pm |H(e^{j\omega})|e^{-j\alpha\omega} \quad (6)$$

Multiplying both sides by $e^{j\alpha\omega}$:

$$\sum_{n=0}^{M-1} h[n]e^{-j\omega(n-\alpha)} = \sum_{n=0}^{M-1} h[n] [\cos(\omega(\alpha - n)) + j \sin(\omega(\alpha - n))] = \pm |H(e^{j\omega})| \quad (7)$$

Since the right-hand side is purely real, the imaginary part must identically equal zero for all ω :

$$\sum_{n=0}^{M-1} h[n] \sin(\omega(\alpha - n)) = 0, \quad \forall \omega \quad (8)$$

This condition can hold for all frequencies ω if and only if:

1. The delay parameter α is exactly the center of symmetry:

$$\alpha = \frac{M-1}{2} \quad (9)$$

2. The impulse response exhibits **even symmetry** about α :

$$h[n] = h[M-1-n], \quad 0 \leq n \leq M-1 \quad (10)$$

4.2 3.2 Condition for Generalized Linear Phase

If we relax the constraint on phase delay and require only constant group delay $\tau_g(\omega) = \alpha$, the phase response is permitted to have a constant offset β :

$$\theta(\omega) = \beta - \alpha\omega \quad (11)$$

Following an identical derivation, the imaginary part condition yields:

$$\beta = \pm \frac{\pi}{2}, \quad \alpha = \frac{M-1}{2} \quad (12)$$

and the impulse response must exhibit **odd symmetry (antisymmetry)** about α :

$$h[n] = -h[M-1-n], \quad 0 \leq n \leq M-1 \quad (13)$$

Critical Consequence for Odd Length M : If M is odd, the center tap index is $n = (M-1)/2$. Substituting this into the antisymmetry condition:

$$h\left[\frac{M-1}{2}\right] = -h\left[M-1 - \frac{M-1}{2}\right] = -h\left[\frac{M-1}{2}\right] \implies h\left[\frac{M-1}{2}\right] = 0 \quad (14)$$

The middle sample of any odd-length antisymmetric FIR filter must always be exactly zero!

5 4. THE FOUR TYPES OF LINEAR-PHASE FIR FILTERS

Based on the symmetry of $h[n]$ and whether the length M is odd or even, linear-phase FIR filters are classified into four distinct types:

Type	Length M	Symmetry of $h[n]$	Group Delay τ_g	Zero at $\omega = 0$?	Zero at $\omega = \pi$?	Allowed Filters
I	Odd	Symmetric ($h[n] = h[M-1-n]$)	$(M-1)/2$ (integer)	No	No	LPF, HPF, BPF, BSF
II	Even	Symmetric ($h[n] = h[M-1-n]$)	$(M-1)/2$ (half-int)	No	Yes (Mandatory)	LPF, BPF only
III	Odd	Antisymmetric ($h[n] = -h[M-1-n]$)	$(M-1)/2$ (integer)	Yes (Mandatory)	Yes (Mandatory)	BPF, Differentiator
IV	Even	Antisymmetric ($h[n] = -h[M-1-n]$)	$(M-1)/2$ (half-int)	Yes (Mandatory)	No	HPF, Differentiator

Table 1: Comprehensive Classification of the Four Types of Linear-Phase FIR Filters.

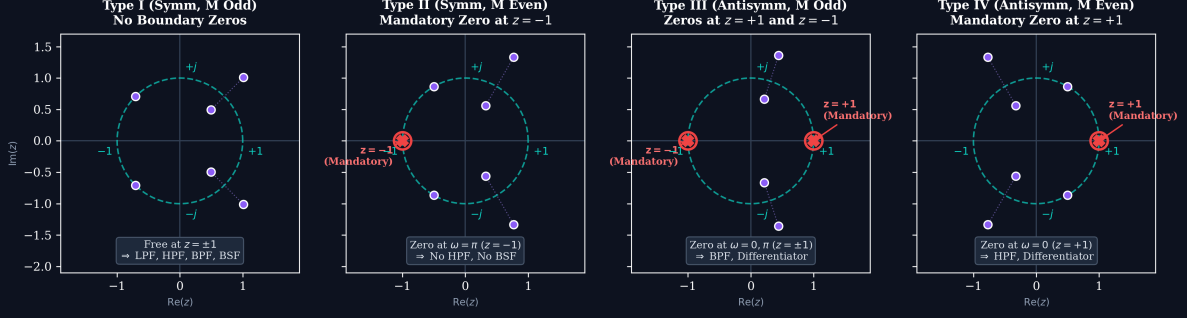


Figure 1: Illustrative z -plane zero constellations reflecting the boundary constraints for the four linear-phase FIR filter types: (a) **Type I**: Free at both boundary points $z = \pm 1$ ($\omega = 0, \pi$), allowing arbitrary low-pass, high-pass, band-pass, and band-stop implementations; (b) **Type II**: Possesses an inherent, unremovable zero at $z = -1$ ($\omega = \pi$), completely ruling out high-pass and band-stop designs; (c) **Type III**: Possesses mandatory zeros at *both* $z = +1$ ($\omega = 0$) and $z = -1$ ($\omega = \pi$), restricting its use to band-pass filters and differentiators; (d) **Type IV**: Possesses a mandatory zero at $z = +1$ ($\omega = 0$), ruling out low-pass designs while making it ideal for high-pass filters and wideband differentiators. Notice that off-axis complex zeros always appear in reciprocal conjugate quadruplets $\{z_0, z_0^*, 1/z_0, 1/z_0^*\}$ to maintain real coefficients and linear phase.

5.1 4.1 Geometric Interpretation of Z-Plane Zero Constellations

The location of zeros in Figure 1 provides profound geometric insight into FIR filter capabilities:

- **Reciprocal Symmetry of Zeros:** For any real-coefficient filter with linear phase, if $z_0 = re^{j\phi}$ is a transmission zero, then:

$$H(z_0) = 0 \implies H(z_0^*) = 0, \quad H(1/z_0) = 0, \quad H(1/z_0^*) = 0 \quad (15)$$

Thus, zeros off the unit circle ($r \neq 1$) and off the real axis must occur in **quadruplets of four zeros**. Zeros on the unit circle ($r = 1$) need only occur in complex conjugate pairs $\{e^{j\phi}, e^{-j\phi}\}$. Real zeros off the unit circle occur in reciprocal pairs $\{r, 1/r\}$.

- **The Boundary Points $z = \pm 1$:** Because $z = 1 = e^{j0}$ corresponds to DC ($\omega = 0$) and $z = -1 = e^{j\pi}$ corresponds to the folding/Nyquist frequency ($\omega = \pi$), a forced zero at either boundary permanently forces the frequency response to zero at that frequency ($H(e^{j0}) = 0$ or $H(e^{j\pi}) = 0$).

5.2 4.2 Frequency Response Expressions

By grouping symmetric pairs, the frequency response can be expressed as a real amplitude function $A(\omega)$ multiplied by a pure phase factor:

$$H(e^{j\omega}) = A(\omega)e^{j(\beta - \alpha\omega)} \quad (16)$$

5.2.1 Type I: Symmetric, Odd Length

Let M be odd. Order $N = M - 1$ is even. Center is at $\alpha = (M - 1)/2$.

$$A(\omega) = h \left[\frac{M-1}{2} \right] + \sum_{k=1}^{(M-1)/2} 2h \left[\frac{M-1}{2} - k \right] \cos(\omega k) \quad (17)$$

Because $\cos(0) = 1$ and $\cos(\pi k) = (-1)^k$, $A(0)$ and $A(\pi)$ are generally non-zero. Hence, Type I can realize any filter type (LPF, HPF, BPF, BSF).

5.2.2 Type II: Symmetric, Even Length

Let M be even. Order $N = M - 1$ is odd. Center is at $\alpha = (M - 1)/2$ (half-integer).

$$A(\omega) = \sum_{k=1}^{M/2} 2h \left[\frac{M}{2} - k \right] \cos \left(\omega \left(k - \frac{1}{2} \right) \right) \quad (18)$$

Evaluating at $\omega = \pi$:

$$A(\pi) = \sum_{k=1}^{M/2} 2h \left[\frac{M}{2} - k \right] \cos \left(\pi k - \frac{\pi}{2} \right) = \sum_{k=1}^{M/2} 2h \left[\frac{M}{2} - k \right] \cdot 0 = 0 \quad (19)$$

Theorem: A Type II filter *always* has an inherent zero at $\omega = \pi$ ($z = -1$). Therefore, **Type II FIR filters can NEVER be used to design High-Pass Filters or Band-Stop Filters!**

6 5. LINEAR-PHASE REALIZATION ARCHITECTURE (FOLDED DIRECT FORM)

6.1 5.1 Standard Direct Form FIR Architecture

A standard transversal FIR filter computes the convolution sum directly:

$$y[n] = \sum_{k=0}^{M-1} h[k]x[n-k] = h[0]x[n] + h[1]x[n-1] + \cdots + h[M-1]x[n-M+1] \quad (20)$$

Hardware requirements for an arbitrary length- M filter:

- **Multipliers:** M
- **Adders:** $M - 1$
- **Delay Elements** (z^{-1}): $M - 1$

6.2 5.2 Exploiting Coefficient Symmetry (Pre-Addition Principle)

Because $h[k] = h[M - 1 - k]$, the multiplier coefficients are pairwise identical! Using the distributive law of arithmetic:

$$h[k]x[n-k] + h[M-1-k]x[n-M+1+k] = h[k] \cdot \left(x[n-k] + x[n-M+1+k] \right) \quad (21)$$

The Architectural Insight: Instead of performing two separate multiplications and then adding their products, we **pre-add** the two symmetric delay taps first, and then perform a **single multiplication!** This cuts the number of required multipliers approximately in half (a **50% reduction in multiplier hardware!**)

6.3 5.3 Exact Realization Equations

6.3.1 Case A: Symmetric, Odd Length (M Odd, Order N Even)

$$y[n] = \sum_{k=0}^{(M-3)/2} h[k] \left(x[n-k] + x[n-M+1+k] \right) + h \left[\frac{M-1}{2} \right] x \left[n - \frac{M-1}{2} \right] \quad (22)$$

- **Multipliers Required:** $\frac{M-1}{2} + 1 = \frac{M+1}{2}$
- **Adders Required:** $\frac{M-1}{2}$ (pre-adders) + $\frac{M-1}{2}$ (accumulation adders) = $M - 1$
- **Delays Required:** $M - 1$

6.3.2 Case B: Symmetric, Even Length (M Even, Order N Odd)

$$y[n] = \sum_{k=0}^{M/2-1} h[k] \left(x[n-k] + x[n-M+1+k] \right) \quad (23)$$

- **Multipliers Required:** $\frac{M}{2}$
- **Adders Required:** $\frac{M}{2}$ (pre-adders) + $(\frac{M}{2} - 1)$ (accumulation adders) = $M - 1$
- **Delays Required:** $M - 1$

6.3.3 Case C: Antisymmetric, Odd Length (M Odd, Center Tap Zero)

$$y[n] = \sum_{k=0}^{(M-3)/2} h[k] \left(x[n-k] - x[n-M+1+k] \right) \quad (24)$$

Because the center coefficient is strictly zero, it requires **no multiplier** at all!

- **Multipliers Required:** $\frac{M-1}{2}$
- **Adders/Subtractors Required:** $\frac{M-1}{2} + \frac{M-3}{2} = M - 2$
- **Delays Required:** $M - 1$

6.3.4 Case D: Antisymmetric, Even Length (M Even)

$$y[n] = \sum_{k=0}^{M/2-1} h[k] \left(x[n-k] - x[n-M+1+k] \right) \quad (25)$$

- **Multipliers Required:** $\frac{M}{2}$
- **Adders/Subtractors Required:** $\frac{M}{2} + (\frac{M}{2} - 1) = M - 1$
- **Delays Required:** $M - 1$

6.4 5.4 Hardware Resource Comparison Table

Filter Length M	Direct Multipliers	Linear-Phase Mults	Multiplier Savings (%)	Total Adders	Total Delays
$M = 4$ (Even, Symm)	4	2	50.0%	3	3
$M = 5$ (Odd, Symm)	5	3	40.0%	4	4
$M = 8$ (Even, Symm)	8	4	50.0%	7	7
$M = 9$ (Odd, Symm)	9	5	44.4%	8	8
$M = 31$ (Odd, Symm)	31	16	48.4%	30	30
$M = 64$ (Even, Symm)	64	32	50.0%	63	63
$M = 128$ (Even, Symm)	128	64	50.0%	127	127

Table 2: Hardware Complexity: Standard Direct Form vs. Linear-Phase Realization.

Key Takeaway for Faculty and Students: Notice that the total number of adders remains **exactly** $M - 1$. Pre-addition does not introduce extra additions into the filter; it merely reorders the additions to occur *before* the multiplications rather than *after* them!

6.5 5.5 Architectural Comparison: Standard Unfolded Cascade of Delays vs. Folded Linear-Phase Structure

To clearly appreciate the hardware transformation, Figure 2 contrasts the standard transversal direct-form structure with the folded linear-phase realization for an $M = 4$ symmetric FIR filter:

1. **(a) Standard Direct-Form (Cascade of Unit Delays):** The input sequence $x[n]$ flows through a straight cascade of $M - 1 = 3$ unit delay elements (z^{-1}). Each tap ($x[n], x[n - 1], x[n - 2], x[n - 3]$) feeds its own dedicated multiplier branch ($h[0], h[1], h[2], h[3]$). An adder chain sums all four products. This requires $M = 4$ full multipliers and $M - 1 = 3$ adders.
2. **(b) Linear-Phase Folded Direct-Form:** Recognizing that $h[3] = h[0]$ and $h[2] = h[1]$, the delay line is folded back on itself. Symmetric taps are aligned vertically:
 - Tap $x[n]$ and tap $x[n - 3]$ are summed by Pre-Adder 1 before multiplying by $h[0]$.
 - Tap $x[n - 1]$ and tap $x[n - 2]$ are summed by Pre-Adder 2 before multiplying by $h[1]$.

Only two products are formed and summed by a single accumulator adder. This preserves exact numerical equivalence while slashing multiplier hardware by 50%!

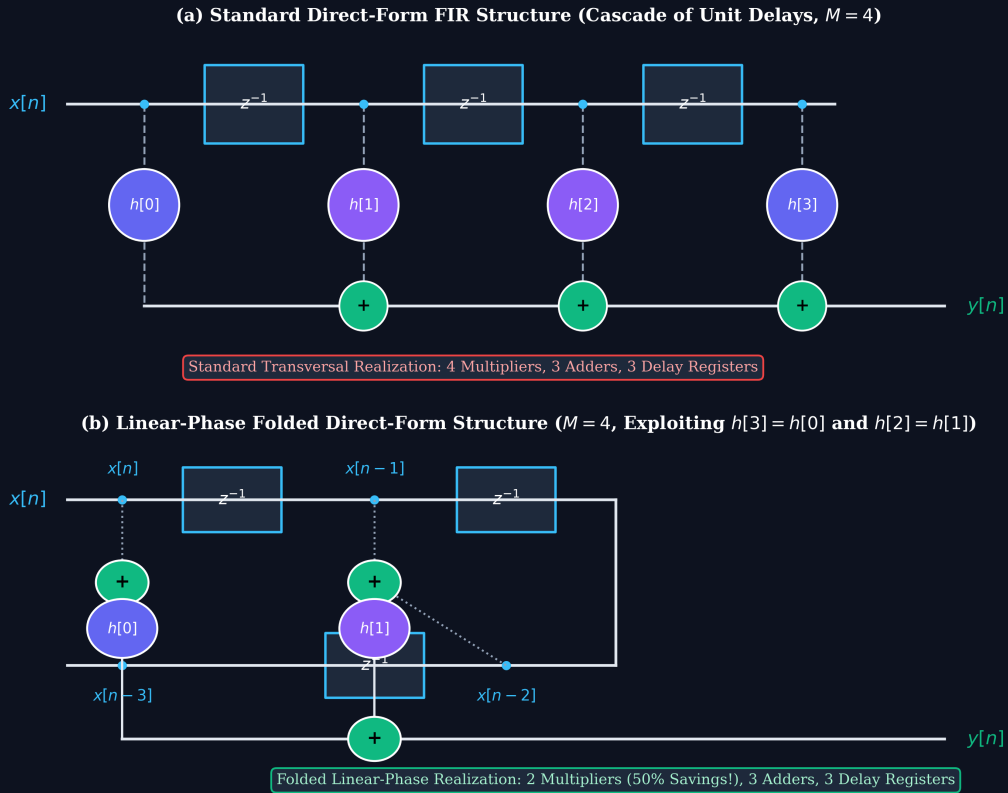


Figure 2: Architectural comparison for $M = 4$ symmetric FIR filter: (a) Standard direct-form structure with a straight cascade of unit delays and 4 separate multipliers; (b) Folded linear-phase structure with pre-adders reducing multiplier count to 2.

Linear Phase FIR Hardware Optimization (50% Multiplier Reduction)

Standard Transversal FIR (M Taps):
Requires M full Multipliers and M-1 Adders

Exploiting Coefficient Symmetry: $h[n] = h[M-1-n]$
Pre-adds symmetric samples: $y[n] = \sum h[k] (x[n-k] + x[n-M+1+k])$
50% Multiplier Savings in Hardware!

Figure 3: Conceptual summary of hardware optimization in linear-phase FIR realizations showing 50% multiplier reduction.

6.6 5.6 Linear-Phase FIR Cascade Realization (Factored Second-Order Sections)

In addition to the folded direct form, linear-phase FIR filters can also be realized in **Cascade Form** by factoring the overall transfer function $H(z)$ into a series connection of lower-order linear-phase sub-filters:

$$H(z) = b_0 \prod_{k=1}^K H_k(z) \quad (26)$$

Because the zeros of any linear-phase filter must exhibit reciprocal symmetry (if z_0 is a zero, then $1/z_0$, z_0^* , and $1/z_0^*$ must also be zeros), each section $H_k(z)$ inherently preserves linear phase:

- **Zeros on the Unit Circle** ($z_k = e^{\pm j\theta_k}$): Form symmetric second-order polynomials:

$$H_k(z) = 1 - 2\cos(\theta_k)z^{-1} + z^{-2} = 1 + \beta_{1k}z^{-1} + z^{-2} \quad (27)$$

Notice that the leading and trailing coefficients are both 1, so each second-order section requires only **one non-trivial multiplier** (β_{1k}) and two adders!

- **Real Zeros** ($z_k = r_k, 1/r_k$): Form symmetric second-order polynomials:

$$H_k(z) = (1 - r_k z^{-1})(1 - r_k^{-1} z^{-1}) = 1 - (r_k + r_k^{-1})z^{-1} + z^{-2} \quad (28)$$

Again, this requires only one non-trivial multiplier.

- **Complex Zeros off the Unit Circle (Quadruplet Zeros)**: Combine into a 4th-order symmetric section with mirrored coefficients: $1 + c_1 z^{-1} + c_2 z^{-2} + c_1 z^{-3} + z^{-4}$.

FIR Cascade-Form realization ($H(z) = H_1(z) \cdot H_2(z)$)

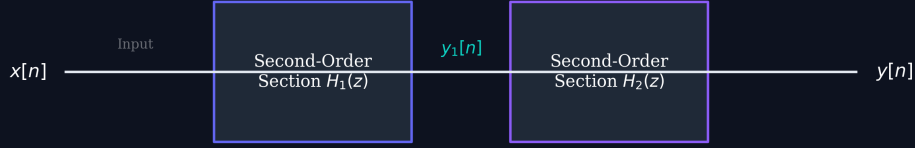


Figure 4: Cascade-form realization of an FIR filter ($H(z) = H_1(z) \cdot H_2(z)$), where each second-order section retains symmetric coefficients to ensure linear phase throughout the cascade.

7 6. VLSI & FPGA IMPLEMENTATION INSIGHTS

7.1 6.1 The Cost of Multipliers in Silicon

In modern digital design (ASIC standard cell libraries and FPGAs):

- An adder is a simple ripple-carry or carry-lookahead circuit requiring very few logic gates ($O(B)$ area for word-length B).
- A multiplier is a 2-dimensional array of full adders or a Wallace-tree structure requiring $O(B^2)$ silicon area and drawing massive dynamic switching power:

$$P_{\text{dynamic}} = \alpha \cdot C_{\text{load}} \cdot V_{DD}^2 \cdot f_{\text{clk}} \quad (29)$$

By eliminating 50% of the multipliers, a linear-phase FIR filter halves the silicon area required for multiplication and drastically cuts the battery consumption of portable DSP devices.

7.2 6.2 Dedicated Pre-Adders in FPGA DSP Slices

Modern FPGAs (such as AMD/Xilinx 7-Series, UltraScale+, and Intel Stratix) contain hard-wired DSP blocks (e.g., DSP48E1 / DSP48E2). Each DSP48E slice contains:

1. A 25-bit **Pre-Adder**.
2. A 25×18 -bit **Two's Complement Multiplier**.
3. A 48-bit **Accumulator / Post-Adder**.

This dedicated architecture evaluates the exact mathematical expression:

$$P = (A + D) \times B + C \quad (30)$$

Because the pre-adder ($A + D$) is physically built right into the DSP slice silicon, **a linear-phase symmetric FIR tap executes in a single clock cycle with zero extra FPGA fabric look-up tables (LUTs)!**

8 7. STEP-BY-STEP WORKED NUMERICAL EXAMPLES

8.1 Example 16.1: Symmetric FIR Filter with Odd Length ($M = 5$)

Problem: An FIR filter has the impulse response:

$$h[n] = \{1, 2, 3, 2, 1\} \quad (31)$$

- (a) Verify that this filter possesses linear phase.
- (b) Write the standard direct form difference equation and state its hardware count.
- (c) Derive the linear-phase direct form difference equation.
- (d) Construct the complete signal flow graph and compare the hardware count.

Solution:

(a) **Verification of Linear Phase:** Here length $M = 5$ (samples indexed $n = 0, 1, 2, 3, 4$).

- $h[0] = 1, h[4] = 1 \implies h[0] = h[4]$
- $h[1] = 2, h[3] = 2 \implies h[1] = h[3]$
- $h[2] = 3$ (center tap, $\alpha = (5 - 1)/2 = 2$)

Since $h[n] = h[4 - n]$ for all $n \in [0, 4]$, the filter is **Type I Symmetric**, exhibiting pure linear phase with constant group delay $\tau_g = 2$ samples.

(b) **Standard Direct Form Difference Equation:**

$$y[n] = 1 \cdot x[n] + 2 \cdot x[n - 1] + 3 \cdot x[n - 2] + 2 \cdot x[n - 3] + 1 \cdot x[n - 4] \quad (32)$$

Hardware resources required:

- Multipliers: 5 (or 3 non-unity, but 5 general multiplier taps).
- Adders: 4.
- Delays: 4 ($z^{-1}, z^{-2}, z^{-3}, z^{-4}$).

(c) **Linear-Phase Form Derivation:** Group terms with identical coefficients:

$$y[n] = 1 \cdot (x[n] + x[n - 4]) + 2 \cdot (x[n - 1] + x[n - 3]) + 3 \cdot x[n - 2] \quad (33)$$

(d) **Hardware Count & Realization Comparison:**

- **Pre-additions:**

$$\begin{aligned} s_0[n] &= x[n] + x[n - 4] \quad (\text{Pre-adder 1}) \\ s_1[n] &= x[n - 1] + x[n - 3] \quad (\text{Pre-adder 2}) \end{aligned}$$

- **Multiplications:** Only 3 multiplications are performed:

$$m_0[n] = 1 \cdot s_0[n], \quad m_1[n] = 2 \cdot s_1[n], \quad m_2[n] = 3 \cdot x[n - 2] \quad (34)$$

- **Accumulation Adders:**

$$y[n] = m_0[n] + m_1[n] + m_2[n] \quad (2 \text{ adders}) \quad (35)$$

- **Total Hardware Audit:**

- Multipliers: $\frac{5+1}{2} = 3$ (Savings: $\frac{5-3}{5} \times 100\% = 40\%$).
- Total Adders: 2 pre-adders + 2 accumulation adders = 4 adders (Unchanged!).
- Delay Registers: 4 delay registers (z^{-1}) (Unchanged!).

8.2 Example 16.2: Symmetric FIR Filter with Even Length ($M = 6$)

Problem: Realize the Type II linear-phase FIR filter having impulse response:

$$h[n] = \{ \underset{\uparrow}{0.5}, -1.2, 2.5, 2.5, -1.2, 0.5 \} \quad (36)$$

in an efficient linear-phase direct form structure.

Solution: Here $M = 6$. Coefficients satisfy:

$$h[0] = h[5] = 0.5, \quad h[1] = h[4] = -1.2, \quad h[2] = h[3] = 2.5 \quad (37)$$

The delay symmetry center is $\alpha = (6 - 1)/2 = 2.5$ samples. Group symmetric input samples:

$$y[n] = 0.5(x[n] + x[n - 5]) - 1.2(x[n - 1] + x[n - 4]) + 2.5(x[n - 2] + x[n - 3]) \quad (38)$$

Hardware Audit:

- **Multipliers:** $\frac{M}{2} = \frac{6}{2} = 3$ multipliers (compared to 6 in direct form $\rightarrow$ **50% hardware reduction!**).
- **Pre-Adders:** 3 adders to compute $(x[n] + x[n - 5])$, $(x[n - 1] + x[n - 4])$, and $(x[n - 2] + x[n - 3])$.
- **Accumulation Adders:** 2 adders to sum the 3 multiplier products.
- **Total Adders:** $3 + 2 = 5$ adders ($M - 1 = 5$).
- **Delays:** 5 delay registers (z^{-1}).

8.3 Example 16.3: Antisymmetric FIR Filter with Odd Length ($M = 5$)

Problem: An odd-length antisymmetric FIR filter is defined by:

$$h[n] = \{ \underset{\uparrow}{1}, 2, 0, -2, -1 \} \quad (39)$$

- Confirm that this is a Type III filter.
- Formulate the efficient realization difference equation.
- Determine the exact number of multipliers and adders.

Solution:

(a) **Classification:** Length $M = 5$ (Odd). Symmetry:

- $h[0] = 1, h[4] = -1 \implies h[0] = -h[4]$
- $h[1] = 2, h[3] = -2 \implies h[1] = -h[3]$
- $h[2] = 0 = -h[2]$ (middle sample is zero!)

This is a **Type III Antisymmetric FIR Filter** with group delay $\tau_g = 2$ samples.

(b) **Difference Equation:**

$$y[n] = 1 \cdot (x[n] - x[n - 4]) + 2 \cdot (x[n - 1] - x[n - 3]) \quad (40)$$

Notice that the term $h[2]x[n - 2] = 0 \cdot x[n - 2] = 0$ is completely absent!

(c) **Hardware Count:**

- **Multipliers:** Exactly $\frac{M-1}{2} = \frac{5-1}{2} = 2$ multipliers! (Compared to 5 in direct form $\rightarrow$ 60% multiplier savings!).
 - **Pre-Subtractors:** 2 subtractors to compute $(x[n] - x[n-4])$ and $(x[n-1] - x[n-3])$.
 - **Accumulation Adders:** 1 adder to sum the two products.
 - **Total Adders/Subtractors:** $2 + 1 = 3$ adders $(M - 2)$.
 - **Delays:** 4 registers (z^{-1}).
-

9 8. PEDAGOGICAL CHECKPOINTS & COMMON MIS-CONCEPTIONS

1. Misconception 1: "Does pre-addition increase the total number of adders in the filter?"

- *Reality:* **No.** The total number of adders is always exactly $M - 1$ for symmetric filters. In standard direct form, all $M - 1$ additions occur in the accumulation bus after multiplication. In linear-phase form, $\lfloor M/2 \rfloor$ additions are shifted before the multipliers, leaving $\lceil M/2 \rceil - 1$ additions for accumulation. The total adder count is mathematically invariant.

2. Misconception 2: "Does the folded linear-phase realization change the transfer function $H(z)$?"

- *Reality:* **No.** The transfer function $H(z)$ and the frequency response $H(e^{j\omega})$ are completely identical. The folded structure is purely an algebraic rearrangement exploiting the distributive law: $a \cdot x_1 + a \cdot x_2 = a(x_1 + x_2)$.

3. Misconception 3: "Can a Type II filter be used to design an audio tweeter high-pass filter?"

- *Reality:* **Absolutely NOT.** A Type II filter (even length, symmetric) has a forced transmission zero at $\omega = \pi$ ($z = -1$), meaning it has zero gain at the highest possible discrete frequency. A high-pass filter must pass $\omega = \pi$. Therefore, attempting to design an HPF using Type II is an immediate fatal design error.

4. Misconception 4: "Can an IIR filter achieve linear phase using this pre-addition technique?"

- *Reality:* **No.** IIR filters have infinite impulse responses that cannot have causal symmetry. Furthermore, IIR realization structures rely on recursive feedback loops ($y[n - k]$ feeding back to $y[n]$), which prevent symmetric pair grouping.
-

10 9. UNIVERSITY EXAMINATION QUESTIONS & PRACTICE PROBLEMS

10.1 Problem 1 (10 Marks — Analytical & Realization)

Question: A linear-phase low-pass FIR filter is specified with impulse response:

$$h[n] = \{-0.1, 0.2, 0.6, 0.6, 0.2, -0.1\} \quad (41)$$

1. State the Type and Order of the filter. (2 Marks)
2. Determine the group delay τ_g and phase delay $\tau_p(\omega)$. (2 Marks)
3. Write the standard direct form and linear-phase direct form difference equations. (3 Marks)
4. Sketch the linear-phase realization structure and evaluate the multiplier savings. (3 Marks)

Marking Scheme & Solution Outline:

1. **Type & Order:** Length $M = 6$, Order $N = M - 1 = 5$. Since $h[n] = h[5 - n]$, it is a **Type II Symmetric FIR Filter**. (2 Marks)
2. **Delays:** $\tau_g = \frac{M-1}{2} = \frac{5}{2} = 2.5$ samples. Phase delay $\tau_p(\omega) = 2.5$ samples (constant). (2 Marks)
3. **Difference Equations:**
 - Direct: $y[n] = -0.1x[n] + 0.2x[n-1] + 0.6x[n-2] + 0.6x[n-3] + 0.2x[n-4] - 0.1x[n-5]$.
 - Linear-Phase: $y[n] = -0.1(x[n] + x[n-5]) + 0.2(x[n-1] + x[n-4]) + 0.6(x[n-2] + x[n-3])$. (3 Marks)
4. **Structure & Savings:** Multipliers reduced from 6 to 3. Savings = 50%. Adders = 5. Delays = 5. (3 Marks)

10.2 Problem 2 (5 Marks — Conceptual Proof)

Question: Prove mathematically that any causal antisymmetric FIR filter of odd length M must satisfy $h\left[\frac{M-1}{2}\right] = 0$. Explain how this property benefits hardware implementation.

Solution Outline: By definition of antisymmetry: $h[n] = -h[M-1-n]$ for $0 \leq n \leq M-1$. Evaluating at the center tap $n = \frac{M-1}{2}$:

$$\begin{aligned} h\left[\frac{M-1}{2}\right] &= -h\left[M-1-\frac{M-1}{2}\right] = -h\left[\frac{M-1}{2}\right] \\ \implies 2h\left[\frac{M-1}{2}\right] &= 0 \implies h\left[\frac{M-1}{2}\right] = 0 \end{aligned}$$

Hardware Benefit: Because the center tap is identically zero, the middle sample $x[n - (M-1)/2]$ requires neither a multiplier nor an adder in the accumulation chain. For an odd-length antisymmetric filter, the required multiplier count drops from M to $\frac{M-1}{2}$ (saving greater than 50%).